



Department of Computer Science and Business Systems
Academic Year 2022 – 2023(Even Semester)

Degree, Semester & Branch: B.Tech., II & CSBS

Course Code & Title: AD3251 Data structures design

Name of the Faculty member: Ms.K.Usharani

Innovative Practice Description

- **Unit / Topic:** Unit 2 / Singly Linked List and Queue ADT
- **Course Outcome:** CO 2 – Design and Implement linear data structures for different applications
- **Topic Learning Outcome:** TLO 6 - Know the basic knowledge on linear data structures like linked list, stack queue and its types.
- **Activity Chosen:** Demonstration
- **Justification:**
 1. Demonstration is a teaching strategy which refers to the practical experimentation of the topic. It'll improve the understanding of the students on the topic given.
 2. Need: Theoretical explanation of concepts is not sufficient and also not understandable one for beginners.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 1. Sources used :
 - a. Virtual Lab - <https://ds1-iiith.vlabs.ac.in/Introduction.html>
 - b. Python IDLE for demonstration
 2. Visualization of data structure is shown with the help of virtual lab.
 3. Experimentation of data structure is done by Lecture with demonstration using python IDLE.
 4. Students also motivated to run the python program in their laptop or in their lab session.

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- **CO – PO / PSO mapping:**

CO	PO1	PO3	PO5	PO12	PSO2
CO5	2	1	3	1	2

(1 – Low

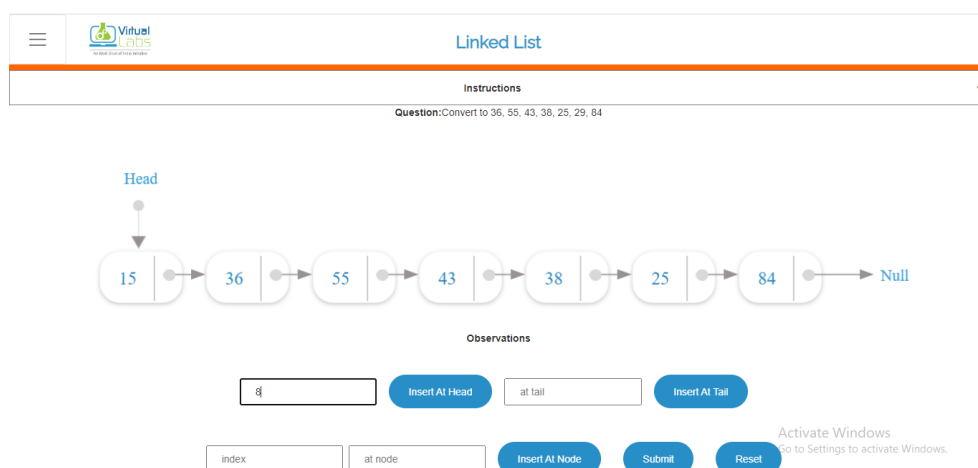
2 – Moderate

3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO3	PO5	PO12	PSO2
	2	1	3	1	2
Justification for correlation	Apply the knowledge of engineering fundamentals.	Design solutions for complex engineering problems	Modern Tool usage like Python IDLE	Recognize the need and have ability to engage in independent and life-long learning.	Exhibit the proficiency in software development.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students felt that Demonstration on the topic gave them a better understanding on the topic. And also gave a hope to work on their own during Lab sessions or at home.

- ❖ ***Benefit of the practice:***

Flipped mode of teaching attracted the student's attention. They gave better focus on the topic than conventional teaching method.

- ❖ ***Challenges faced in implementation:***

It is not a student centered method. Students cannot actively participate in his process. The ultimate responsibility of carrying out the experiment lies with the teacher. But the students are motivated to do on their own.

References:

1. <https://ds1-iiith.vlabs.ac.in/Introduction.html>